

IMAGE PROCESSING AND ITS APPLICATION (IPCC)			
Course Code	22RI63	CIE Marks	50
Teaching Hours/Week (L:T:P)	3-0-2	SEE Marks	50
Total Hours of Pedagogy	52	Total Marks	100
Credits	04	Exam Hours	03
Course objectives: • To understand the basics of image processing • To examine the working of filters and morphological operations • To examine the working of neural networks • To analyze various deep learning algorithms • To analyze YOLO algorithm and other neural networks			
Course Outcomes: At the end of the course student will be able to CO1: Understand the basics of image processing CO2: Use different filters and to perform morphological operations CO3: Apply neural networks for various operations using open CV CO4: Analyse various deep learning algorithms CO5: Analyse YOLO algorithm and other neural networks			
Module-1			
Introduction to Image Processing: Introduction to Image Processing, Fundamental steps in image processing, Components of Image Processing system, Image sensing and acquisition, Image sampling and quantization, Application of Image processing, Color Models			CO1
Module-2			
Image Processing Techniques: Basic Grey level transforms, Histogram processing, histogram equalization, Smoothing frequency domain filters, Sharpening frequency domain filters, Image degradation/restoration process, noise models, Mean filter, Median filter, Mid-point filter, Image segmentation, Morphological operations			CO2
Module-3			
Introduction to open CV and neural networks: Image processing using open CV, Shape detection using open CV, Fundamentals of Deep learning, Layers in neural network, neurons, hyperparameters, bias, activation function, sigmoid function, tanh function, ReLU, Softmax function, learning rate, backpropagation, overfitting, gradient descent, loss function, deep learning libraries, Tensor, convolution, Pooling layer, fully connected layer			CO3

Module-4		
Introduction to object detection:		CO4
LeNet – 1, LeNet – 4, Boosted LeNet – 4, LeNet – 5 architecture, AlexNet neural network, VGG Neural network – VGG 16, Object detection and use cases, Object detection methods, Deep learning frameworks for object detection, R- CNN architectures		
Module-5		
Introduction to deep learning algorithms:		CO5
Features and Loss function in YOLO algorithm, YOLO architecture, Single shot multibox detector – SSD, Transfer learning, FaceNet for face recognition, ResNet architecture, Inception network, GoogLeNet architecture, Video processing		
COURSE CONTENTS (PRACTICAL)		
#	NAME OF THE EXPERIMENT	CO
1	Write a program to perform basic image processing operations	1
2	Write a program to generate the histogram of an image and filter the image	2
3	Write a program to perform image segmentation and morphological operations	2
4	Write a program to detect color using open CV	3
5	Write a program to detect shapes using open CV	3
6	Write a program to detect face using open CV	3
7	Write a program to distinguish between cat and dog based on their image	3
8	Write a program to classify models using LeNet	4
9	Write a program to identify traffic signs using LeNet	4
10	Write a program to implement FaceNet for face recognition	5
11	Write a program to recognize gestures using python	5
12	Write a program to detect multiple objects in an image	5
Suggested Learning Resources:		
Books		
1. Divya BS, Nanjundaswamy HR, “Image processing” (Module 1 and 2)		
2. Vaibhav Verdhan “Computer vision using deep learning” apress publication (Module 3,4,5)		
3. Digital Image Processing by Rafael C. Gonzalez		
Reference Books		
6. Fundamentals of Digital Image Processing by Anil K. Jain		
7. Image Processing and Acquisition using Python by Ravishankar Chityala		
8. Introduction to Digital Image Processing by William K. Pratt		

CO-PO Map

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO															
CO 1	3	2	2		3				3	2		2		3	
CO 2	3	3	3		3				3	2		2		3	
CO 3	3	3	3		3				3	2		2		3	
CO 4	3	3	3		3				3	2		2		3	
CO 5	3	3	3		3				3	2		2		3	

3 – Strong, 2 Moderate, 1-Weak